Strategic renunciation of AI and digital technologies in local contexts: feasibility, risks, and alternatives

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Abstract

The uncontrolled and swift evolution of Artificial Intelligence, with the emphasis on efficiency at the cost of societal impact, demands substantial resources. Techno-critics and environmental activists are deeply concerned with the pace of these advances, and there's not a single framework to evaluate the actual consumption of data centers. These considerations lead to the idea of replacing heavy AI models with AI-reduced alternatives in local contexts. In this study, the feasibility of inference localization was evaluated on light-weight models by processing question categorization and subject recommendation tasks in the university context. Performance analysis was carried out concerning several optimizations such as batching, parallelization, and quantization. Some techniques, including neural classifiers and transformers, were additionally executed on a GPU for measuring acceleration. Sentence embeddings appeared to be the most balanced in terms of trade-offs between accuracy, computational cost, and memory consumption. Rule-based systems functioned better for precise tasks where semantics were less crucial. Generally, heavier models such as DistilBERT and Tinyllama performed worse in a narrow environment due to insufficient training. The results suggest that for constrained tasks, advanced generative models might not be necessary, because smaller models may achieve great precision as well at a tiny fraction of the original cost. However, these solutions usually lack standardization and require more manual work.

1 Introduction: Research Problem and Objectives

1.1 Background

Over the past decade, Artificial Intelligence (AI) has been developing rapidly, and many business sectors, including the Information and Communication Technology (ICT) sector, have become increasingly dependent on it. AI systems strongly rely on high-performance hardware such as graphics processing units (GPUs) and tensor processing units (TPUs). Deep learning architectures now require substantial computational resources to achieve high performance, and the needed allocation grows expeditiously [Goodfellow *et al.*, 2016]. Similarly, massive architectures known as warehouse-scale computers have been described as integrated systems in which data centers function as a single computational unit [Barroso *et al.*, 2013].

All modern AI models go through two main phases: training and inference. Training consumes more resources because it adjusts billions of model parameters using large datasets and boosting algorithms such as gradient descent. Extensive matrix operations and parallel computation are essential for this phase, making it computationally intense. On the contrary, more lightweight inference aims to generate predictions on an already trained model, and it can sometimes even be executed locally.

However, modern large language models still require significant hardware resources, especially when deployed at scale. Some researchers are concerned that increasing the accuracy of these models requires larger computational resources, leading to higher consumption of energy, water, and other resources. They propose reporting training time and sensitivity to hyperparameters, which will allow to assess the compatibility of the provided resources with the current setting before reusing the same models [Strubell *et al.*, 2019].

Other researchers address the gap in measuring AI's carbon emissions, energy, and water consumption in the production environment by proposing a comprehensive approach, including accelerator, central processing unit (CPU), and random access memory (RAM) energy, idle energy, and overhead energy for cooling and power conversion. They also report that by following this methodology, the actual resource consumption during inference is one or two orders of magnitude smaller than many existing estimates [Elsworth *et al.*, 2025].

1.2 Motivation

Centralization of AI systems, which is specifically crucial for training, leads to an increase in efficiency through batching and other optimization techniques, but it comes at the cost of concentration of computational power and additional infrastructural dependence. The techno-critical movement broadly discusses the environmental implications of such infrastructures. While the ICT sector is estimated to produce 2-4% of all carbon emissions, the boom of infrastructure-intensive facilities correlates with the increased consumption of water and electricity. For instance, Google's data center in Dalles, Oregon, uses more than 350 gallons of water, which is merely a third of the total water use in the city, and more than 330 kWh of electricity, more than 2/3 of the total use in the district. Plans to expand the already existing network of 8000 data centers outweigh the idea of consuming less network data. The climate impact of the latter is negligible due to the fixed high use of power by datacenter equipment [Pasek, 2023].

Furthermore, Bender *et al.* [2021] claimed that increasingly large language models become more and more widespread, arguing that their environmental and societal damage is disregarded relative to their benefits.

While most research focuses on improving AI performance and scalability, limited attention has been given to the feasibility and implications of reducing or renouncing AI usage in specific contexts, especially from a technical perspective. Then, the question arises: to what extent is it technically and organizationally achievable to reduce AI usage in a controlled environment such as a university without significantly degrading system performance and usability? And more precisely, to what extent can inference tasks be decentralized or localized by using lighter and non-AI alternatives? We propose several hypotheses:

H1: Lightweight AI models provide most of the utility of large models at a fraction of the computational cost.

H2: For narrowly defined university tasks, non-generative approaches can achieve acceptable performance; there's no need for state-of-the-art AI.

H3: Optimization techniques such as quantization, batching, and parallelization substantially reduce computational requirements.

This report examines the university model case study and evaluates the practicality of replacing heavy cloud AI-based systems with more lightweight AI-reduced counterparts. This paper evaluates infrastructure measurements such as memory usage, inference and training time, CPU and GPU allocation utilized by common tasks, for example, university-related questions categorization and semester subject recommender. This paper also presents the possible compromises in controlled settings: local language learning models (LLMs), sentence transformers and embeddings, local neural classifiers, text vectorizers, and rule-based systems. This work may challenge the overusage of the most resource-intensive technologies, such as generative AI, and provide insights into more sustainable, lower-complexity practices that preserve local autonomy and accuracy.

2 State of the Art

2.1 AI Infrastructure

AI systems strongly rely on high-performance hardware such as GPUs and tensor processing units (TPUs). While central processing units (CPUs) are optimized for general sequential computation, they're outperformed in neural training by GPUs and TPUs that contain thousands of smaller cores designed for parallel computation. Deep learning models rely on matrix multiplications and tensor operations; consequently, the growth of computation requirements has moved AI development to accelerators such as GPUs and TPUs. This hardware shift explains why AI infrastructure is concentrated in vast datacenters owned by tech giants like Google, Microsoft and Amazon.

At the same time, advanced AI models are too large for a single computer. Training deep learning models, especially in natural language processing or computer vision, entails distributed simultaneous computation across several machines, sometimes involving entire data centers. Frameworks such as DistBelief divide datasets and distribute calculations between clusters of servers or processors. Asynchronous stochastic gradient descent (Downpour SGD) is one of the optimizations that allows multiple machines to update model parameters in parallel. Distributed training reduces the training time with trade-offs of a significant rise in energy consumption, synchronization overhead, fault tolerance, and reliance on centralized infrastructure. Consequently, the structural reliance on large-scale systems is stressed to the extent that even local applications are dependent on remote computational resources [Dean *et al.*, 2012].

To boost hardware utilization, especially on GPUs, AI systems process batches of requests during training and inference. Processing many user requests simultaneously can cause higher efficiency in production environments, as it's suggested by Google's environmental impact study. It also implies that industrial-scale deployment may be more energy-efficient per request than isolated local computation. However, batching also fosters centralization because efficiency is directly connected to scaling and aggregation of workloads.

From an architectural standpoint, data centers contain multiple layers with the aim of performance optimization, reliability, and scalability. Basically, single servers are combined into racks, which in turn connect through a rapid network to create clusters. Orchestration software is in charge of these clusters, and it considers available resources to divide computational tasks between them. Data is accessed and processed in parallel, and storage systems are also distributed. The entire data center functions as an individual entity rather than a group of independent machines. This design got the name "warehouse-scale computer". While this model is highly efficient and fault-tolerant, it also presents challenges related to resource allocation and cooling, which require a lot of water, signaling the complexity and cost of maintaining AI infrastructure.

2.2 Environmental and Legal Context

Some researchers argue that the inflating demands of data centers can be compensated for by following best practices:

choosing sparse model architectures, using optimized processors, computing in the cloud, and picking a location for a data center with the cleanest energy. Fostering competition between companies by publishing carbon footprint, energy usage, and Power Usage Effectiveness (PUE), combined with purchasing renewable energy to match the consumption, may yield great results. For instance, after the introduction of the Transformer model in 2017, in just 4 years, its training CO2 emissions were 747 times minimized [Patterson *et al.*, 2022].

From the legal point of view, data center restrictions exist on different levels, including the EU Artificial Intelligence Act, which promotes energy efficiency and sustainability. With international policies and local laws such as the Amsterdam Duurzaam Digital, the Netherlands managed to use more than 85% of the data centers' electricity from renewable sources.

2.3 Alternatives to heavyweight AI

Not all AI requires massive infrastructure; for instance, tiny Machine Learning and edge AI execute machine learning models on low-power devices like microcontrollers [Han *et al.*, 2026]. Local data processing leads to advantages such as lower latency and energy cost, reduced network reliance, and improved privacy. But these models suffer from limited model complexity and a lack of common deployment approaches among different devices [Alajlan and Ibrahim, 2022].

Partial decentralization is technically feasible by Term Frequency-Inverse Document Frequency (TF-IDF) retrieval and sentence embeddings. The first method statistically represents documents by ranking the importance of words within collections of text. The second method better understands the context as it converts text into vectors that capture semantics, allowing to compare the similarity between sentences and text.

Another low-computation technique is rule-based systems. They make decisions based on pre-defined rules, which are transparent and don't need training. While being inexpensive, they're the least flexible as they tend to overfit the data.

3 Methodology

3.1 University tasks

To evaluate the AI-reduced university model, two different tasks were proposed. The first one was a categorization of frequently asked questions with 30 questions and 10 categories. Only a few questions were ambiguous; for instance, the question "Can international students get housing?" could belong to both "housing" and "mobility" categories. If the training was involved, the model was also trained on 30 pairs of questions and categories. The expected output for each question was the most well-suited category.

The second task was less direct: recommendation of courses for the semester. Among 14 possible subjects, depending on the user query involving preferences, the models were expected to output the 5 best and 5 worst subject matches. Subject descriptions were retrieved from the course presentations. Instead of ranking subjects, only the rule-based system was requested to recommend the most suitable subject because of the method's limitations.

We experimented on tiny datasets that represented narrow university scenarios. Therefore, measurements, including accuracy, were not indicative of large-scale deployment but were demonstrative of relative differences between methods. For higher precision, all methods were run 5 times, and their mean measurements were considered stable.

3.2 Compared methods

We selected these methods for comparison:

- **Rule-based systems** - very low-cost non-AI approach functioning on manually defined conditions, doesn't require training.
- **TF-IDF** - non-AI approach processing documents depending on term importance, similarity is rather statistical than semantic, and requires minimal training.
- **Neural classifier** - a lightweight supervised classifier trained on labeled institutional data, serves as a small AI.
- **Sentence Transformer** - AI-based encoding text into semantic embeddings, includes semantic similarity search, and doesn't require training.
- **DistilBERT** - compressed transformer trained through the knowledge distillation, a more lightweight version of Bidirectional Encoder Representations from Transformers.
- **Tinyllama** - a tiny generative language model, this local cloud-independent LLM was pre-trained.

3.3 Measured metrics

We evaluated different approaches based on these metrics:

- **Accuracy** - for the categorization task, it's a percentage of correctly assigned categories to questions. For the recommendation task, we evaluated how similar final course placements with subjectively pre-defined labels were. If the subject was correctly predicted to be suitable or non-suitable, 1 point was given, and if it was placed incorrectly (recommendation of a non-suitable subject or vice versa), it was deducted from the final score, after that the score was normalized.
- **Inference time** - time for solely making predictions.
- **Training time** - time to process the training set and set up the model.
- **RAM** - difference between RAM after and before inference or training.
- **GPU memory** - evaluated on T4 GPU in Google Colab
- **CPU utilization** - the most fluctuating metric, taken as an average of 1 second.

3.4 Optimization techniques

For boosting the performance, we utilized batching of sizes of 1, 10, 30 for the categorization task and sizes of 1, 5, 15 for the recommendation task. We also utilized parallelization in

neural classifiers, sentence embeddings, and DistilBERT for concurrent inference, which can be extremely beneficial on GPUs. Running many repetitive mathematical operations on several levels reduces training time, but this method heavily depends on specialized infrastructure. We also used quantization in DistilBERT, Tinyllama Q4 (quantized version of Tinyllama); this technique reduces precision from floats to integers, resulting in smaller models and faster CPU inference. We also considered pruning, but removing unimportant weights was not practical because we didn't train deep networks from scratch.

4 Results

4.1 Memory consumption and accuracy

We considered how RAM usage correlates with the accuracy, and the results are presented in Figure 1. Bigger memory consumption was generally correlated with worse accuracy: rule-based systems, sentence transformers, and neural classifiers were more accurate than relatively heavy DistilBERT.

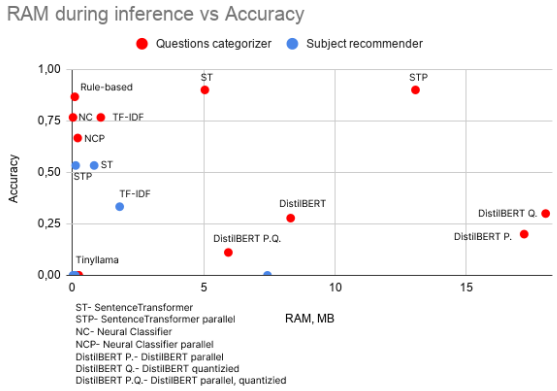


Figure 1: Comparison of RAM usage and accuracy across the evaluated methods for categorization and recommendation tasks.

Also, all models had higher accuracy for the question categorizer than for the subject recommender: for the latter, the maximum accuracy achieved by sentence transformers is 53%, while for the categorizer, half of the models achieved an accuracy higher than 67%, with the maximum of 90% for sentence transformers. For all versions of DistilBERT, accuracy was lower than 27%, and Tinyllama didn't generate a proper output to evaluate, so its accuracy was estimated as null.

Parallelization didn't change the accuracy for sentence transformers, but increased RAM usage. However, parallelized DistilBERT was incorrect more often than its sequential counterpart; the parallelized neural classifier also had degraded precision.

A similar trend can be observed when we measured memory usage on the GPU. Figure 2 summarizes the calculations. For example, a rule-based approach and TF-IDF used negligible memory while achieving an accuracy above 75% for categorization. Sentence transformers and neural classifiers were more memory-hungry. DistilBERT was times more memory-intensive, while the parallelized version had worse accuracy and also consumed the most memory.

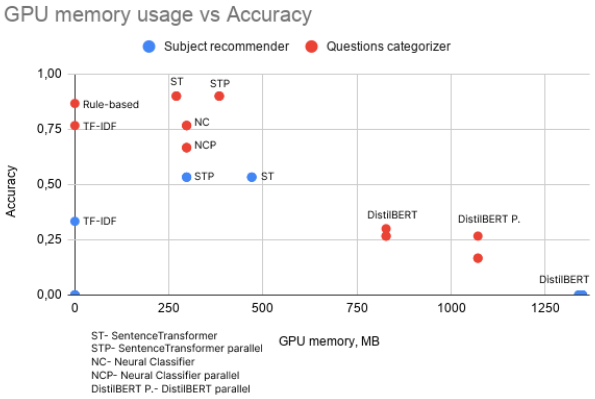


Figure 2: Comparison of GPU memory usage and accuracy across the evaluated methods for categorization and recommendation tasks.

4.2 Computational cost

Figure 3 presents the inference time for the categorization task across various models. While Tinyllama requires more than 100 seconds for inference, other models need less than a second, and rule-based systems need 4 orders of magnitude less time. This is consistent with the previous findings, suggesting that generative AI is by far the most time-consuming in terms of inference time.

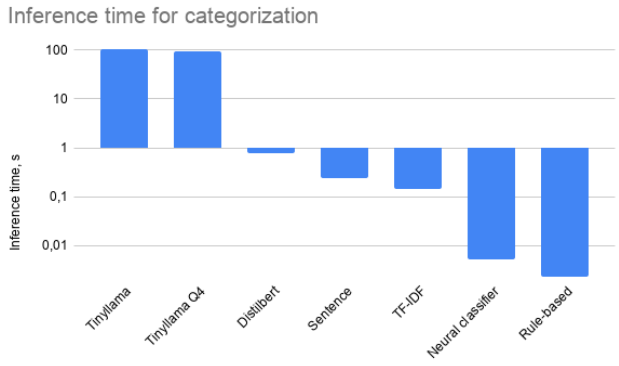


Figure 3: Comparison of inference time across the evaluated methods for categorization.

4.3 Inference time and accuracy

Figure 4 demonstrates the link between accuracy and inference time. Low-cost models presented better performance: for instance, rule-based systems took around 2 ms for inference while providing 87% accuracy for categorization. On the contrary, poorly performing Tinyllama needed around 100

s for inference, and more than 500 s for the recommender task. Both quantization and parallelization reduced inference time: parallelized neural classifier and sentence embeddings were approximately 2 times faster than their sequential versions. For categorization, quantized and parallelized DistilBERT received the inference boost of 5 times.

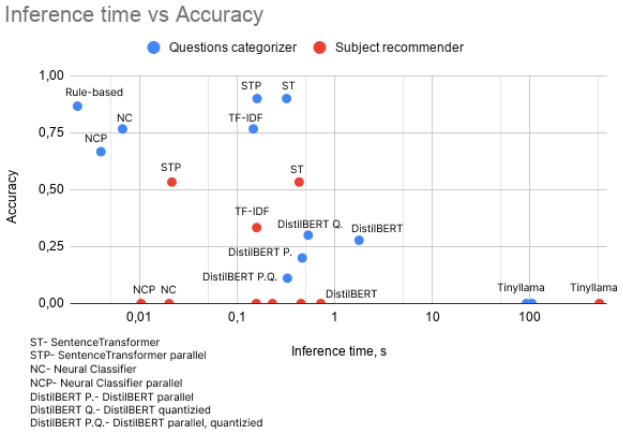


Figure 4: Comparison of inference time and accuracy across the evaluated methods for categorization and recommendation tasks.

4.4 Training and inference time comparison

Figure 5 shows how much time training and inference require. For both tasks, we observed that training is more costly than inference in terms of time for the neural classifier and DistilBERT. While training usually took 15-45× times longer, for subject recommendation, it took 326× times longer, because training on large texts was more intense.



Figure 5: Comparison of training and inference time for neural classifiers and DistilBERT performing categorization and recommendation tasks.

4.5 Impact of optimizations

Batching produced speedups between 1.5× and 3× for transformer-based methods, with the neural classifier benefiting most (up to 17×). Figure 6 presents the acceleration effect of batching for sentence transformers. For sentence embeddings, inference time gradually reduced with the bigger batches, on average 2-2,5 times. The only exception is the parallel version of the subject recommender - inference time was already reaching the model limit without batching, so it stayed constantly low after the grouping.

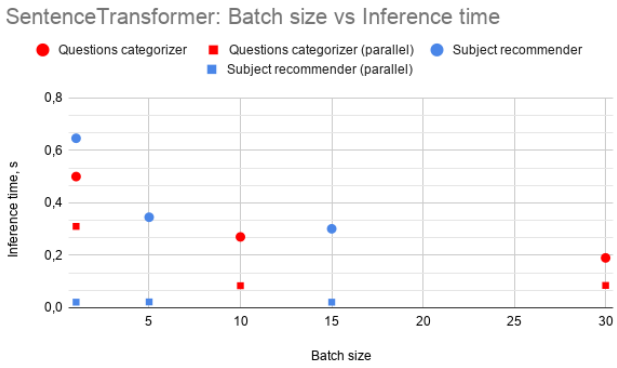


Figure 6: Comparison of batch sizes and inference time for sentence transformers performing categorization and recommendation tasks.

Table 1 summarizes the average speedups of optimizations. On GPUs, sentence transformers achieved inference acceleration of around 38-65%. With more heavyweight DistilBERT, we can observe that training can speed up more tremendously (22-23 times) than inference (7-8 times) by using GPUs instead of CPUs. Quantization reduced DistilBERT inference times by approximately 2-6×, while parallelization did so by about 35-50%.

Table 1: Acceleration of various optimizations

Optimization	Average speedup
Batching	2-3×
Parallelization	1.5×
Quantization	3×
GPU	4-6×

5 Discussion

The main finding is that small systems are surprisingly competitive. For both local institutional tasks, large language models didn't automatically outperform simpler methods. We can assume that for university applications, sentence embeddings, rule-based approaches, and TF-IDF may be enough. They placed first, second, and third, respectively, in average accuracy (if we consider 40% accuracy in a single recommender task for rule-based systems), while requiring fewer computational resources.

Sentence transformers performed exceptionally well. One possible explanation is that semantic similarity is sufficient for university tasks. This supports the idea that some AI use cases in local contexts can be successfully replaced with

lower-complexity options; hypotheses H1 and H2 were confirmed.

Clearly, training is far more costly than inference, specifically for larger models like DistilBERT. The experiments showed that the main computation burden is training, as inference can be executed even on consumer hardware. Both stages can benefit from using GPUs, even though this acceleration is more tangible for larger models.

We originally assumed in hypothesis H3 that optimization techniques could minimize computation requirements. The experiments confirmed that batching improved inference throughput by sharing computational costs between multiple requests. Quantization was the most consistent optimization; it substantially reduced inference costs while maintaining stable performance. Dynamic quantization primarily improved execution speed while not reducing the utilized memory that much. Parallelization produced mixed results. The effect of parallelization was limited for small workloads because coordination overhead prevailed over computational gains. A combination of these techniques produced better performance metrics but not necessarily better accuracy.

DistilBERT and Tinyllama underperformed in a constrained environment. General-purpose LLMs incurred dramatically higher inference costs without providing good precision. A tiny dataset of 30 training examples and insufficient fine-tuning degraded DistilBERT performance. We acknowledge that transformer models require substantially larger datasets for better precision, so the obtained accuracy is not an architecture limitation but the consequence of data scarcity.

The poor performance of Tinyllama can be explained by assuming that general-purpose LLMs can be inefficient for narrow tasks, and long queries are hard to process. Unlike the other methods, TinyLLama was not specifically trained for the evaluated tasks and therefore based its reasoning only on the prompts.

For subjective tasks, rule-based systems become weaker than for tasks with fixed categories. It provides evidence that semantic similarity is more crucial for recommendation than categorization, so this approach might be less universal outside structured tasks.

These findings suggest that AI renunciation should not necessarily be interpreted as the complete removal of automated systems. When task complexity is low and doesn't justify additional infrastructure requirements, partial renunciation can be implemented by replacing costly AI with lighter models.

6 Conclusion

AI reduction does not mean abandoning automation, which could be difficult. It should not be viewed as a binary choice between using and abandoning AI. It means choosing lighter alternatives and the redistribution of labor. Instead, different levels of computational complexity can be selected according to task requirements.

For constrained university applications, such as frequently asked questions categorization and course recommendation, lightweight approaches, including rule-based systems, TF-IDF retrieval, and sentence embeddings, achieved competitive results while requiring considerably less computational resources than large language models. These models consistently provided the best balance between accuracy, cost, and infrastructure requirements. Local approaches can also be considered as infrastructurally resilient, as they don't rely on the internet as Cloud LLMs do. These findings indicate that partial AI reduction may be technically feasible in local contexts without a substantial loss of functionality.

For future work on this topic, I envision applying optimization techniques to training with larger datasets - specifically, batching can make a positive impact on performance. More advanced measurements on GPUs and possibly TPUs, including temperature and processor fluctuations, can also be advantageous for future research. Local context needs to be more clearly defined and extrapolated not only in the simulated environment, but also in the real-life university deployment. Some methods could still achieve great precision, but scalability and maintenance are two of the biggest concerns.

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